

# Polynomial Regression

Machine Learning Assignment

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GitHub Repository: [github.com/TheOrangeCarXX/polynomial\\_regression](https://github.com/TheOrangeCarXX/polynomial_regression)

## 1 Introduction

This assignment focuses on predicting a continuous target variable  $y$  using polynomial regression on two personalized datasets, **var1** and **var2**. Each dataset contains different numbers of input variables and has a different maximum permitted polynomial degree.

For **var1**, six input variables ( $x_1, \dots, x_6$ ) are provided, with a maximum polynomial degree of 10. For **var2**, three input variables ( $x_1, x_2, x_3$ ) are provided, with a maximum polynomial degree of 20.

The objective is to select an appropriate polynomial degree using cross-validation, train the final model on the complete training data, and generate predictions for the provided test datasets.

The primary evaluation metrics are Mean Squared Error (MSE) and the coefficient of determination ( $R^2$ ).

## 2 Methodology

The same general workflow is followed for both problems:

1. Load the training dataset and separate input variables  $X$  from the target  $y$ .
2. Generate polynomial features up to the candidate degree.
3. Evaluate polynomial degrees 1 through the maximum using 5-fold cross-validation with Ridge regression.
4. Select the degree that minimises the mean validation MSE.
5. Retrain using RidgeCV over a fine logarithmic grid of regularization parameters on the complete training set.
6. Apply the trained pipeline to the test dataset to generate predictions.
7. Save the required prediction CSV file.

Polynomial features are generated as total-degree expansions. For three variables and degree two, the feature expansion contains

$$1, x_1, x_2, x_3, x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2.$$

A degree- $d$  expansion includes all monomials for which the sum of exponents is at most  $d$ .

All features are standardized prior to regression. Ridge regularization is used throughout to control the effect of the large number of polynomial features, especially at higher degrees. The regularization parameter  $\alpha$  is selected automatically by RidgeCV using cross-validation.

The implementation uses a `scikit-learn` Pipeline combining `PolynomialFeatures`, `StandardScaler`, and `RidgeCV`, with `n_jobs=-1` for parallel cross-validation.

## 3 Problem 1: var1

### 3.1 Dataset

The **var1** dataset is associated with the problem of *Power Plant Steam Turbine Optimization*. It contains six input variables,

$$x_1, x_2, x_3, x_4, x_5, x_6,$$

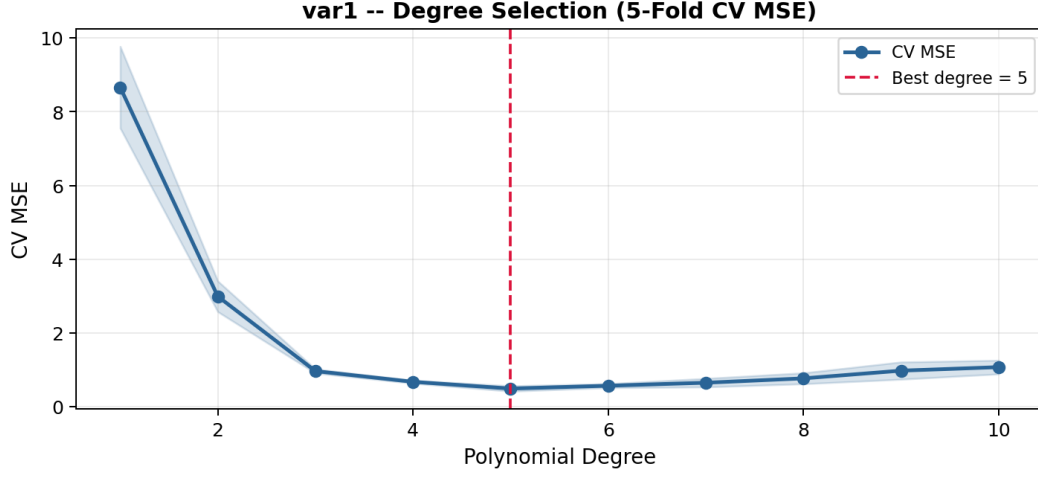
and the target variable  $y$ . Both the training and test datasets contain 1000 samples each. The target variable  $y$  is available in the training dataset but withheld from the test dataset.

The maximum permitted polynomial degree for this problem is 10.

### 3.2 Degree Selection

Polynomial degrees 1 through 10 were evaluated using 5-fold cross-validation. For each degree, a full pipeline of polynomial feature expansion, standard scaling, and Ridge regression (`RidgeCV`) was fitted independently on each fold.

Figure 1 shows the mean validation MSE together with a  $\pm 1$  standard-deviation band across the five folds.



**Figure 1:** 5-fold cross-validation MSE for polynomial degrees 1–10 on the `var1` dataset. The dashed red line marks the selected degree.

The cross-validation MSE decreases sharply from degree 1 to degree 5 and then levels off or increases slightly at higher degrees, indicating overfitting. **Degree 5** is selected as the optimal degree for `var1`.

### 3.3 Final Model

The final `var1` model configuration is:

- Polynomial degree: 5
- Regression method: Ridge (`RidgeCV`)
- Regularization parameter  $\alpha = 24.4205$ : selected automatically by `RidgeCV` over a 50-point logarithmic grid in  $[10^{-4}, 10^4]$
- Feature standardization: `StandardScaler`
- Total polynomial terms: 462

The final model is retrained on the complete training dataset (1000 samples). Its training-set performance is:

| Metric         | Value   |
|----------------|---------|
| Training MSE   | 0.20328 |
| Training $R^2$ | 0.97912 |

**Table 1:** Training performance of the final `var1` model.

These figures describe performance on the training data and should not be interpreted as the hidden test-set evaluation.

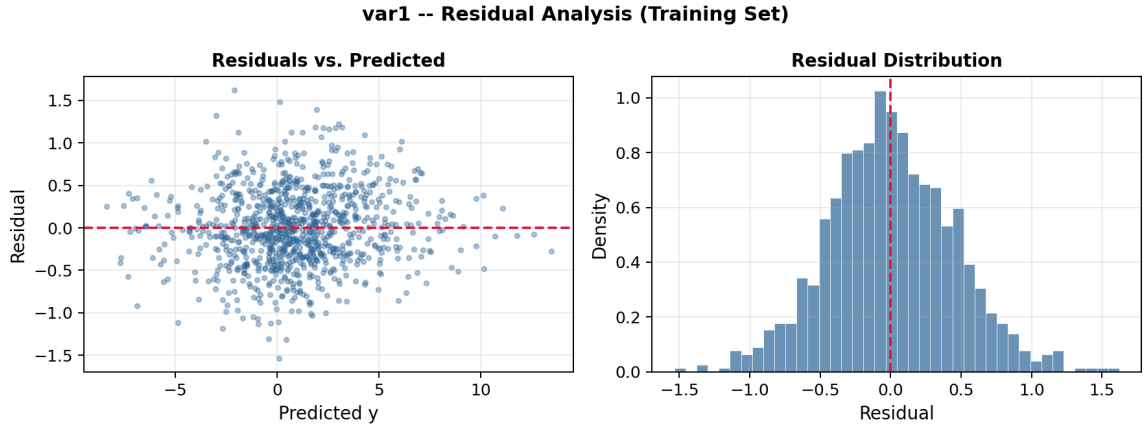
For reference, the cross-validation results across all evaluated degrees for `var1` are:

| Polynomial Degree | CV MSE         |
|-------------------|----------------|
| 1                 | 8.66342        |
| 2                 | 2.99288        |
| 3                 | 0.96692        |
| 4                 | 0.67628        |
| 5                 | <b>0.49331</b> |
| 6                 | 0.57071        |
| 7                 | 0.65166        |
| 8                 | 0.76998        |
| 9                 | 0.97906        |
| 10                | 1.07415        |

**Table 2:** 5-fold cross-validation MSE for all polynomial degrees evaluated on **var1**. The bold entry indicates the selected degree.

### 3.4 Residual Analysis

Figure 2 presents the residual analysis for the final **var1** model.



**Figure 2:** Residual analysis for the final **var1** model. *Left:* residuals plotted against predicted values. *Right:* distribution of residuals.

The residuals are spread approximately symmetrically around zero with no pronounced structure, which is consistent with a well-specified model.

## 4 Problem 2: var2

### 4.1 Dataset

The **var2** dataset is associated with the problem of *Subterranean Thermal Reservoir Mapping*. It contains three input variables,

$$x_1, x_2, x_3,$$

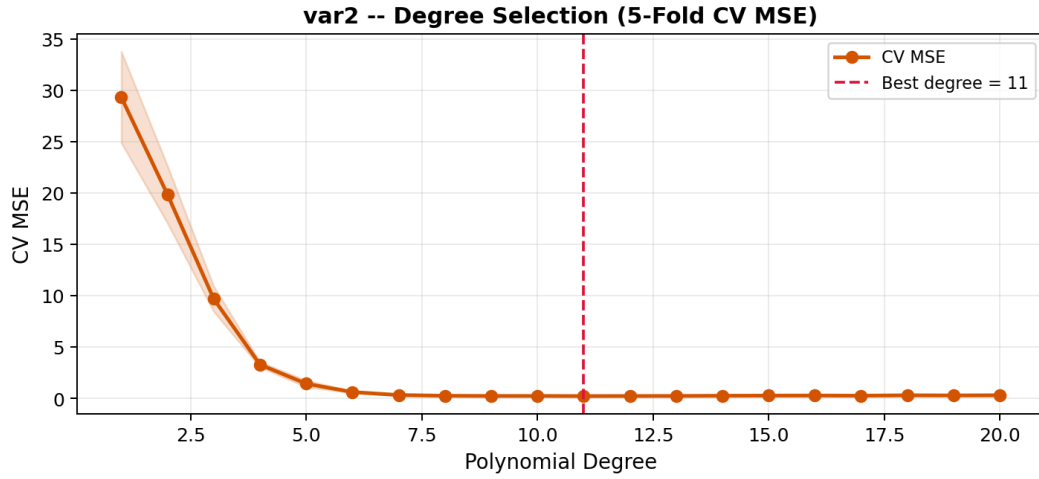
and the target variable  $y$ . Both the training and test datasets contain 1000 samples each. The target variable is available only in the training dataset.

The maximum permitted polynomial degree for this problem is 20.

### 4.2 Degree Selection

Polynomial degrees 1 through 20 were evaluated using 5-fold cross-validation with the same Ridge pipeline used for **var1**.

Figure 3 shows the mean validation MSE and the corresponding  $\pm 1$  standard-deviation band.



**Figure 3:** 5-fold cross-validation MSE for polynomial degrees 1–20 on the `var2` dataset. The dashed red line marks the selected degree.

The validation MSE decreases rapidly for low degrees and then stabilises, reaching a minimum at degree 11. Higher degrees show no further improvement. The cross-validation results for a selection of degrees are shown below.

| Polynomial Degree | CV MSE         | CV $R^2$       |
|-------------------|----------------|----------------|
| 1                 | 29.34065       | 0.26060        |
| 2                 | 19.85724       | 0.49514        |
| 3                 | 9.69979        | 0.75272        |
| 4                 | 3.30961        | 0.91375        |
| 5                 | 1.44844        | 0.96324        |
| 6                 | 0.63411        | 0.98391        |
| 7                 | 0.34300        | 0.99128        |
| 8                 | 0.27327        | 0.99297        |
| 9                 | 0.25543        | 0.99341        |
| 10                | 0.25587        | 0.99343        |
| 11                | <b>0.24284</b> | <b>0.99372</b> |
| 12                | 0.25031        | 0.99355        |
| 13                | 0.25714        | 0.99339        |
| 14                | 0.27629        | 0.99295        |
| 15                | 0.28879        | 0.99266        |
| 16                | 0.29618        | 0.99250        |
| 17                | 0.28042        | 0.99284        |
| 18                | 0.31837        | 0.99198        |
| 19                | 0.30358        | 0.99229        |
| 20                | 0.32038        | 0.99189        |

**Table 3:** 5-fold cross-validation results for all polynomial degrees evaluated on `var2`. Degree 11 achieves the lowest CV MSE and is selected for the final model.

**Degree 11** achieves the lowest validation MSE (0.24284) among all evaluated degrees and is therefore selected for the final `var2` model.

### 4.3 Final Model

The final `var2` model configuration is:

- Polynomial degree: 11
- Regression method: Ridge (`RidgeCV`)
- Regularization parameter  $\alpha = 1.2068$ : selected automatically by `RidgeCV` over a 50-point logarithmic grid in  $[10^{-4}, 10^4]$
- Feature standardization: `StandardScaler`
- Total polynomial terms: 364

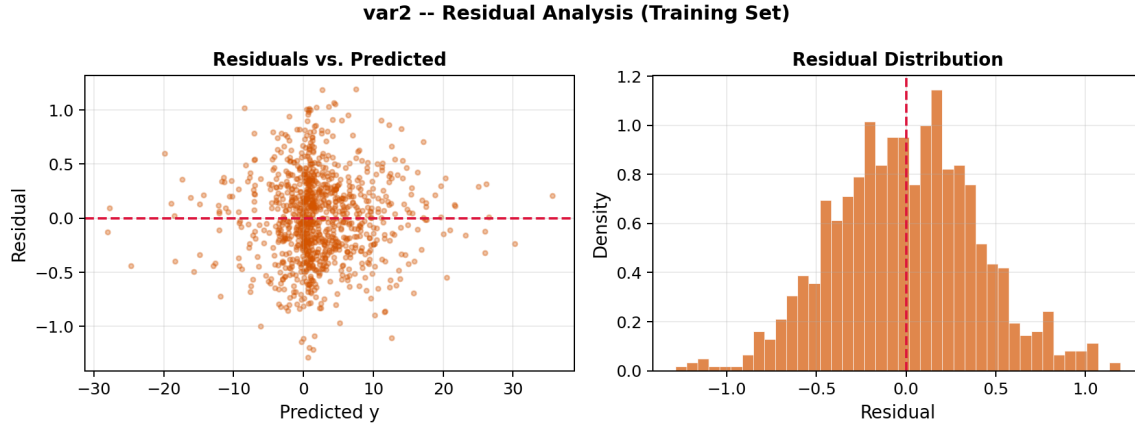
The final model is retrained on the complete training dataset. Its training-set performance is:

| Metric         | Value   |
|----------------|---------|
| Training MSE   | 0.16019 |
| Training $R^2$ | 0.99602 |

**Table 4:** Training performance of the final `var2` model.

## 4.4 Residual Analysis

Figure 4 shows the residual behavior of the final degree-11 Ridge model on the training set.



**Figure 4:** Residual analysis for the final `var2` model. *Left:* residuals plotted against predicted values. *Right:* distribution of residuals.

The residuals are centered near zero with no evident pattern, and their distribution is approximately symmetric, which is consistent with a well-fitted model.

## 5 Inference and Prediction

After degree selection and final model training, the trained pipelines are applied to the corresponding test datasets:

$$X_{\text{test}} \longrightarrow \text{Polynomial Feature Expansion} \longrightarrow \text{StandardScaler} \longrightarrow \text{Trained Ridge Model} \longrightarrow \hat{y}_{\text{test}}.$$

Because the true target values of the test datasets are withheld, the test-set MSE and  $R^2$  cannot be computed locally.

The final prediction files are:

- `BT2024103_pred_var1.csv`
- `BT2024103_pred_var2.csv`

Each file contains a single column `y` with the predicted values for the corresponding test dataset (1000 rows each).

## 6 Implementation

The entire solution is contained in a single script, `ml_assignment_solution.py`, organized into the following logical stages:

- **Data loading:** reads the four CSV files (`train_var1`, `test_var1`, `train_var2`, `test_var2`).
- **Degree selection** (`find_best_degree`): iterates over candidate degrees, builds the Ridge pipeline, and evaluates it with 5-fold CV using both MSE and  $R^2$  scoring.
- **Final model training** (`train_final_model`): refits the pipeline on the full training set using a fine logarithmic grid of  $\alpha$  values via `RidgeCV`.
- **Prediction & saving:** applies each trained pipeline to the corresponding test features and saves the results as CSV.
- **Visualization:** generates the degree-selection and residual-analysis figures used in this report.

All computations use `scikit-learn` pipelines to ensure that feature scaling is estimated only on the training fold during cross-validation, preventing data leakage.

## 7 Conclusion

Polynomial regression was applied to both personalized datasets using a unified Ridge regularization pipeline and 5-fold cross-validation for degree selection.

For **var1** (Power Plant Steam Turbine Optimization), polynomial degree 5 was selected. The final degree-5 Ridge model ( $\alpha = 24.4205$ ) achieved a training MSE of 0.20328 and a training  $R^2$  of 0.97912, with 462 polynomial features.

For **var2** (Subterranean Thermal Reservoir Mapping), polynomial degree 11 was selected based on the lowest validation MSE (CV MSE = 0.24284, CV  $R^2 = 0.99372$ ) across degrees 1–20. The final degree-11 Ridge model ( $\alpha = 1.2068$ ) achieved a training MSE of 0.16019 and a training  $R^2$  of 0.99602, with 364 polynomial features.

The trained models were applied to the corresponding test datasets, producing the two required prediction CSV files (`BT2024103_pred_var1.csv` and `BT2024103_pred_var2.csv`).